



Bachelor study programme
Bioinformatics and Computational Biology
A-4232 Hagenberg, Austria

Identification of Modifications in Open Searches

Bachelor thesis

to obtain the academic degree of
Bachelor of Science in Engineering

Submitted by

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Hagenberg, September 2026

Declaration

I hereby declare and confirm that this thesis is entirely the result of my own original work. Where other sources of information have been used, they have been indicated as such and properly acknowledged. I further declare that this or similar work has not been submitted for credit elsewhere. This printed copy is identical to the submitted electronic version.

Date

Signature

Acknowledgment

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Abstract

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Kurzfassung

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Chapter 1

Introduction

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Chapter 2

Fundamentals and state of the art

Proteomics, MS/MS, PSMs, open search, PTMs, localisation, Unimod, related tools

2.1 Biological Background: Proteins and Modifications

2.2 Proteomics and Mass Spectrometry

2.3 Database search and open modification searches

2.4 Modification annotation and localisation

2.5 Modification databases and related tools

2.6 Implications for the proposed tool

Chapter 3

Concept/Design

Requirements, architecture, data model, algorithm, ranking model, error cases

Chapter 4

Implementation, tests and analyses

Implementation, input/output, testing, datasets, evaluation, results, limitations

Chapter 5

Conclusion

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5.1 Future Work

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Appendix A

Demo Chapter

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Appendix B

Source Codes

List of source codes:

B.1 Snippets

```
int main() {  
    return 0;  
}
```

Program B.1: Small example code

Table B.1: List of abbreviations
Appendix C

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
BMBF	Bundesministerium für Bildung und Forschung

References

- An, Z., Zhai, L., Ying, W., Qian, X., Gong, F., Tan, M., & Fu, Y. (2019). PTMiner: Localization and Quality Control of Protein Modifications Detected in an Open Search and Its Application to Comprehensive Post-translational Modification Characterization in Human Proteome*. *Molecular & Cellular Proteomics*, 18(2), 391–405. <https://doi.org/10.1074/mcp.RA118.000812>
- Jiang, Y., Rex, D. A. B., Schuster, D., Neely, B. A., Rosano, G. L., Volkmar, N., Momenzadeh, A., Peters-Clarke, T. M., Egbert, S. B., Kreimer, S., Doud, E. H., Crook, O. M., Yadav, A. K., Vanuopadath, M., Hegeman, A. D., Mayta, M. L., Duboff, A. G., Riley, N. M., Moritz, R. L., & Meyer, J. G. (2024). Comprehensive Overview of Bottom-Up Proteomics Using Mass Spectrometry. *ACS Measurement Science Au*, 4(4), 338–417. <https://doi.org/10.1021/acsmeasuresciau.3c00068>

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